

XAI for Heart Sound Analysis

RMS Review 2 — Phase 1: Data Preprocessing Pipeline

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Guide: Dr. Sapan Mankad • Course: RMS

Review 1 Recap

Dataset

Preprocessing Pipeline

Feature Extraction & DataLoader

Next Steps

Review 1 Recap

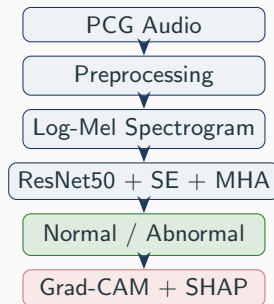
Review 1 — What We Proposed

Literature Review

- Surveyed **5 papers**: heart sound classification + XAI
- Reference: *Alrabie & Barnawi, 2025*

Proposed Architecture

- **ResNet50** — pretrained backbone
- **SE Block** — channel attention
- **Multi-Head Attention** — temporal context
- **Grad-CAM + SHAP** — explainability



Dataset

Dataset — PhysioNet CinC 2016

- **3,541** recordings — training + validation
- Binary labels: Normal vs Abnormal
- Subsets: training-a to training-f + validation
- Variable lengths: **5 s – 120 s**

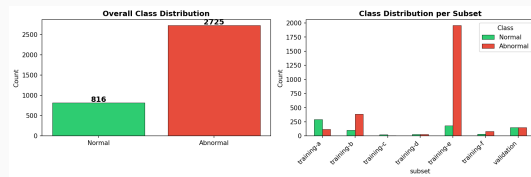


Figure 1: Class distribution — overall and per subset

Class Imbalance

Normal	816 (23%)
Abnormal	2,725 (77%)

Handled via **class weights** in loss function.

Preprocessing Pipeline

Pipeline Overview

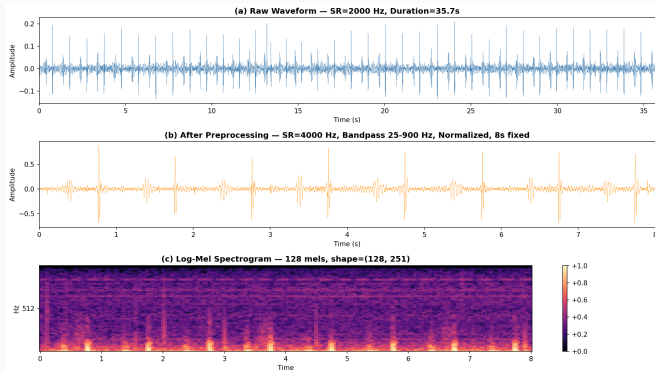
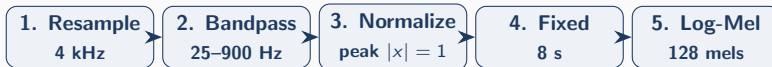


Figure 2: Raw waveform → filtered signal → Log-Mel spectrogram

Key Parameters

SR	4 kHz
Bandpass	25-900 Hz
Length	8 s
Mels	128
Shape	(128, 251)

Output saved as `.npy` for fast loading.

Bandpass Filter Design

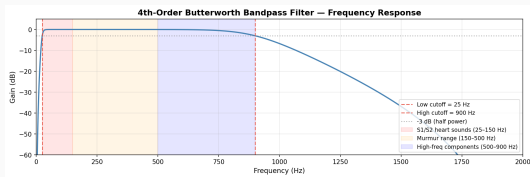


Figure 3: 4th-order Butterworth — 25 to 900 Hz

Bands Preserved

- 25–150 Hz — S1/S2 heart sounds
- 150–500 Hz — Murmur energy
- 500–900 Hz — High-freq detail

- Butterworth: flat passband, no ripple
- Removes baseline drift (< 25 Hz)
- Removes high-freq noise (> 900 Hz)

Why 8 Seconds? — Duration Analysis

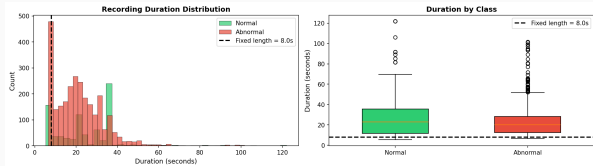


Figure 4: Recording duration distribution by class

- Most recordings: **5–40 s**
- **8 s** covers ≥ 8 heartbeats at normal HR
- Below median of both classes
- Short clips: **zero-padded**
- Long clips: **cropped from start**
- Uniform tensor shape for batching

S1/S2 Detection — Hilbert Envelope

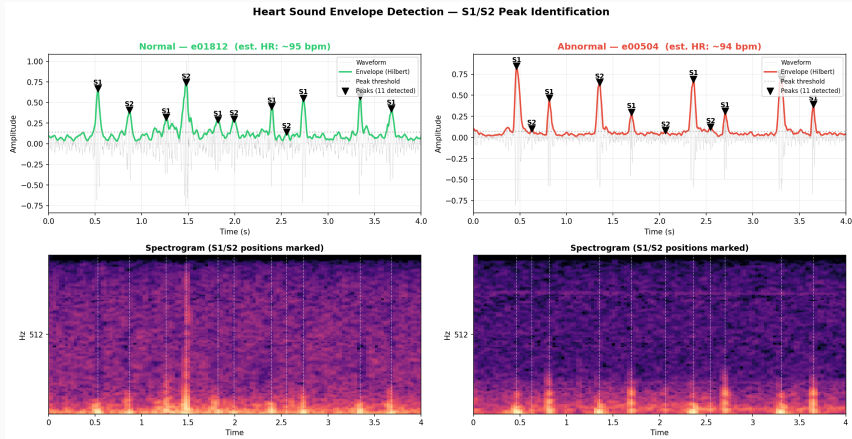


Figure 5: Envelope + peak detection: Normal (left) and Abnormal (right). HR $\approx$ 94–95 bpm.

- Hilbert transform $\rightarrow$ analytic envelope
- Peaks above threshold tagged S1/S2

- Consistent HR validates preprocessing
- *Analysis only* — not a training feature

Feature Extraction & DataLoader

Log-Mel Spectrograms & DataLoader

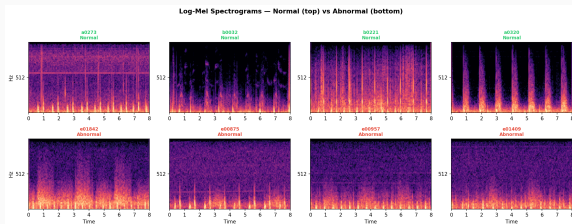


Figure 6: **Normal** (top row) vs **Abnormal** (bottom row)

Spectrogram Config

Mel bins	128
Shape	(128, 251)
Saved as	.npz

DataLoader Config

Resize	224×224
Channels	3 (RGB)
Normalise	ImageNet μ, σ
Augment	SpecAugment
Split	70 / 15 / 15
Batch size	32
Imbalance	Class weights

SpecAugment — Data Augmentation

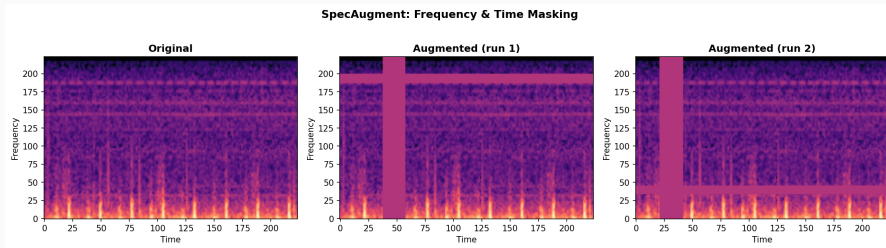


Figure 7: Original (left) vs two SpecAugment runs — random frequency and time masks applied

- Randomly zeros out contiguous frequency bands and time steps
- Forces model to be robust to partial occlusions
- Applied *only during training*

Next Steps

Roadmap — Phases 2 to 6

Phase 2 — Model

ResNet50 + SE Block + Multi-Head Attention

Phase 3 — Training

Frozen backbone → full fine-tune
Weighted BCE + label smoothing

Phase 4 — Evaluation

Accuracy · F1-Score · AUC-ROC

Phase 5 — XAI

- **Grad-CAM**: spatial saliency on spectrogram
- **SHAP**: feature importance per prediction

Phase 6 — Comparison

Benchmark vs Alrabie & Barnawi (2025)
and other surveyed literature

Summary

Summary

- ✓ Dataset explored — PhysioNet CinC 2016, 3,541 recordings, imbalance identified
- ✓ 5-step pipeline: resample → bandpass → normalise → fixed-length → spectrogram
- ✓ Design choices validated: duration analysis, filter response, S1/S2 detection
- ✓ PyTorch DataLoader: SpecAugment, class weights, stratified 70/15/15 split
- ✓ 128-mel spectrograms saved as .npy — ready for model training

References

References

- Clifford et al., *CinC*, 2016
- Alrabie & Barnawi, *BSPC*, 2025
- He et al., *CVPR*, 2016
- Hu et al., *CVPR*, 2018
- Vaswani et al., *NeurIPS*, 2017
- Selvaraju et al., *ICCV*, 2017
- Lundberg & Lee, *NeurIPS*, 2017
- Park et al., *Interspeech*, 2019

Thank You

Questions?